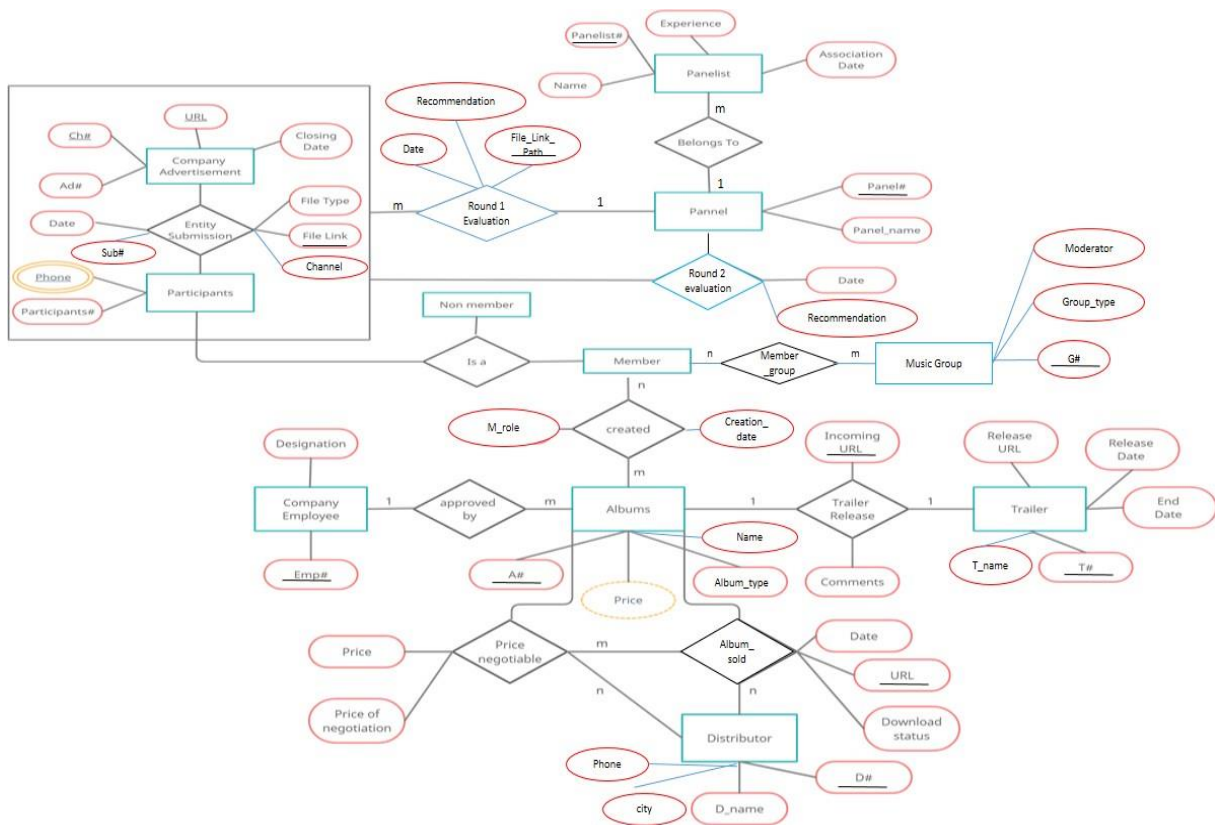


Solution for Tutorial-1 and Tutorial-2

Question1



Design and draw an ER diagram. List your assumptions, if any. We will accept all reasonable assumptions. Your E-R diagram should clearly --

- Identify all entities and their attributes
- Identify relationships between these entities and their attributes
- Identify and underline the primary key of each entity
- Identify entity relationship participation and constraints

Entities-

Company_Advt (Advt#, Channel, URL, Starting_Date, Closing_Date)

Participants (Participant#, Participant_Name, {Phone#}, City, State

Note: Phone is multi-valued

Panellist (Pannelist#, Name, Years_Experience, Joining_Date, Panel#)

Panel (Panel#, Panel_Name)

Company_Employee (E#, Desgination)

Albums (A#, Name, App_E#, Album_type, Price)

Music_Group (Group#, Name, Group_type, Moderator#)

Trailer (T#, T_Name, R-URL, R-Date, E-Date)

Distributor (D#, D_Name, Phone, City)

- Channel and media file can be weak entity.

Relationships with attributes

Entry_Submission (Participant#, URL, Channel, File_Type, Submission_Date, File_Link_Path, S#)

Round1_evaluation (Panel#, File_Link_Path, Date, Recommendation)

Round2_evaluation (Panel#, Member#, Recommendation, Date)

Album_creation (A#, M#, M_Role, Creation_Date)

Tariler_Release (A#, T#, Incoming_URL, Comments)

Album_sold (A#, D#, URL, Date)

Member_Group (M#, Group#)

Price_negotiable(A#, D#, Negotiated_Price, Date)

Question 2: Convert your E-R diagram of Question 1 into relational schemas.

Company_Advt (Advt#, Channel, URL, Starting_Date, Closing_Date)

Participants (Participant#, Participant_Name, {Phone#}, City, State)

Entry_Submission (Participant#, URL, Channel, File_Type, Submission_Date, File_Link_Path, S#)

Panellist (Pannellist#, Name, Years_Experience, Joining_Date, Panel#)

Panel (Panel#, Panel_Name)

Round1_evaluation (Panel#, File_Link_Path, Date, Recommendation)

Round2_evaluation (Panel#, Member#, Recommendation, Date)

Company_Employee (E#, Desgination)

Albums (A#, Name, App_E#, Album_type, Price)

Album_creation (A#, M#, M_Role, Creation_Date)

Music_Group (Group#, Name, Group_type, Moderator#)

Member_Group (M#, Group#)

Trailer (T#, T_Name, R-URL, R-Date, E-Date)

Tariler_Release (A#, T#, Incoming_URL, Comments)

Distributor (D#, D_Name, Phone, City)

Album_sold (A#, D#, URL, Date, Download status)

Price_negotiable(A#, D#, Negotiated_Price, Date)

Question 3: Write the Relational Algebraic Expression (RAE) for each of the following query on relational schemas of Question 2.

A. List all 'Audio' albums released in 2020.

$\sigma_{album_type=Audio \wedge creation_date(year)=2020}(Album)$

B. List all participants who have submitted both Audio and Video files.

$\pi_{Participants\#}(\sigma_{FileType=Audio \wedge A \wedge FileType=Video}(Entry_submission))$

C. List all members who have been the member of more than one group

$\pi_{memberName}(Member_Group - (\pi_{memberID}(Member_Group)))$

D. List all members of 'Pop' music group who are not part of any other music group.

$\pi_{memberID}(\sigma_{Group_Type="pop"}(Member_Group \bowtie Music_group)) -$

$\pi_{memberID}(\sigma_{Group_Type \neq "pop"}(Member_Group \bowtie Music_group))$

E. List all distributors who sold all types of albums

$\pi_{d\#}(\sigma_{AlbumType=Audio}(Album_sold \bowtie Album)) \cap$

$\pi_{d\#}(\sigma_{AlbumType=video}(Album_sold \bowtie Album))$

Question 4: Write the SQL statement on relational schemas of Question 2.

- A. List all 'Audio' albums released in 2020.

```
SELECT Album#  
FROM Album  
WHERE albumType =Audio AND creation_date(year) =2020;
```

- B. List all members who have been the member of more than one group.

```
SELECT M#  
FROM Member_Group  
WHERE ( SELECT Distinct member as m  
        FROM Music_Group  
        WHERE Member_Group.m = m# AND  
        GROUP By group_name)>1;
```

- C. List all members of 'Pop' music group who are not part of any other music group.

```
SELECT DISTINCT Member_Group.m#  
FROM Member_Group, Music_group  
WHERE Music_group.G# = Member_Group.G# AND  
      Music_group.Group_type = 'POP' AND  
      NOT EXIST IN  
      (SELECT M#  
       FROM Member_Group  
       WHERE Music_group.Group_type != 'POP');
```

- D. List all participants who have submitted both Audio and Video files.

```
SELECT participant#  
  
FROM Entry_submission  
WHERE fileType =Audio AND fileType =video;
```

- E. The McM company would like to analyze the data for "Which advertisement channel has been effective that attracted maximum number of entry submissions?"

```
SELECT channel, COUNT(*)  
FROM Entry_submission  
GROUP BY channel;
```